

Design a Relational Database for Smart Toys Co.

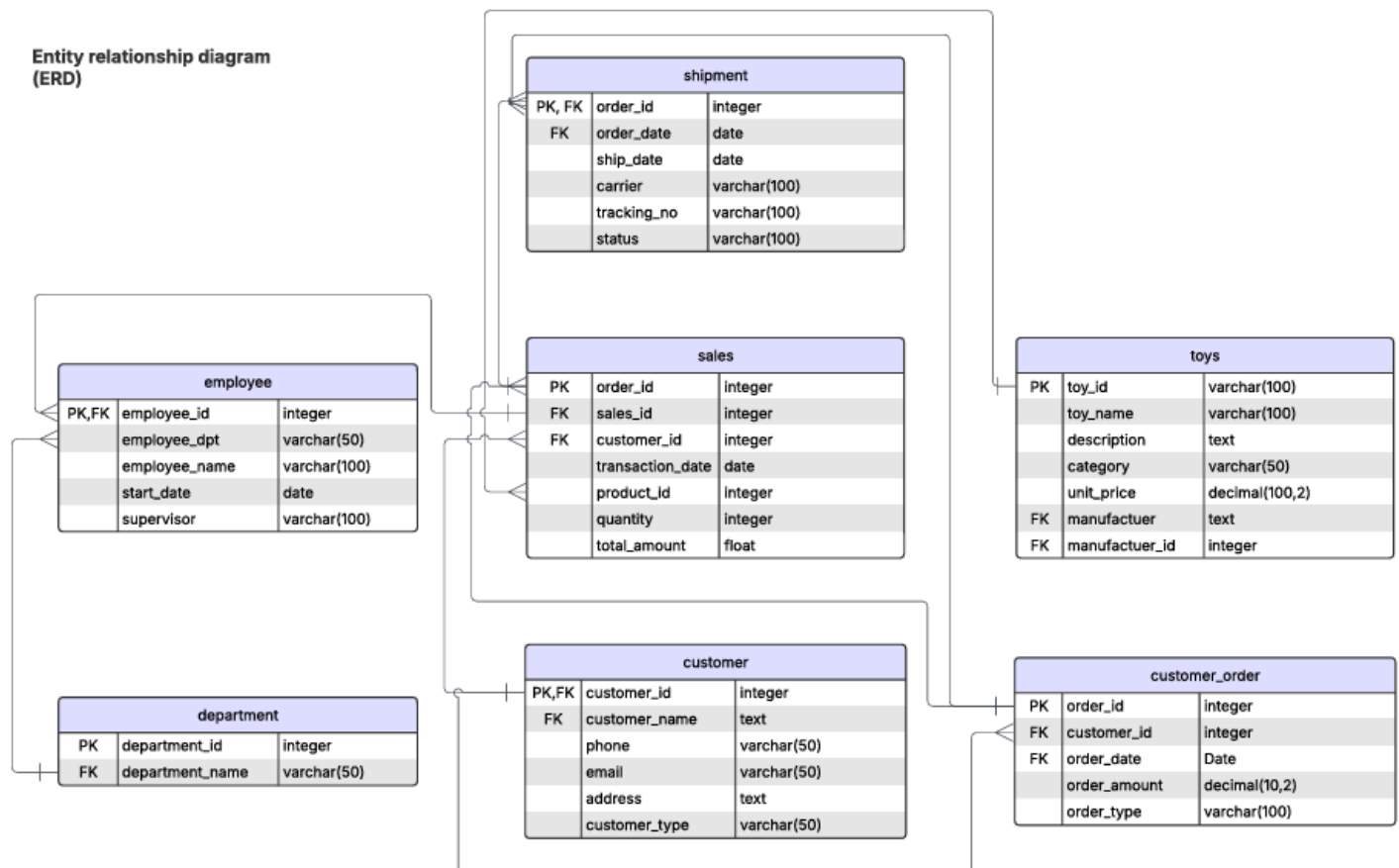
This project consists of 3 parts, including:

Part I: Design the database schema that defines the key entities and their relationships.

Part II: Construct a relational database and tables.

Part III: Write queries to answer business questions.

Part I: Design the Database Schema



Part II: Create a Relational Database

```
CREATE TABLE `customer` (  
  `customer_id` integer,  
  `customer_name` text,  
  `phone` varchar(50),  
  `email` varchar(50),  
  `address` text,  
  `customer_type` varchar(50),  
  PRIMARY KEY (`customer_id`)  
);
```

```
CREATE TABLE `customer_order` (  
  `order_id` integer,  
  `customer_id` integer,  
  `order_date` Date,  
  `order_amount` decimal(10,2),  
  `order_type` varchar(100),  
  PRIMARY KEY (`order_id`),  
  FOREIGN KEY (`customer_id`)  
    REFERENCES `customer` (`customer_id`)  
);
```

```
CREATE TABLE `toys` (  
  `toy_id` varchar(100),  
  `toy_name` varchar(100),  
  `description` text,  
  `category` varchar(50),  
  `unit_price` decimal(100,2),  
  `manufactuer` text,  
  `manufactuer_id` integer,  
  PRIMARY KEY (`toy_id`)  
);
```

```
CREATE TABLE `sales` (  
  `order_id` integer,  
  `sales_id` integer,  
  `customer_id` integer,  
  `transaction_date` date,  
  `product_id` integer,  
  `quantity` integer,  
  `total_amount` float,  
  PRIMARY KEY (`order_id`),  
  FOREIGN KEY (`product_id`)  
    REFERENCES `toys` (`toy_id`),  
  FOREIGN KEY (`customer_id`)  
    REFERENCES `customer` (`customer_id`),  
  FOREIGN KEY (`order_id`)  
    REFERENCES `customer_order` (`order_id`)  
);
```

```
CREATE TABLE `shipment` (  
  `order_id` integer,  
  `order_date` date,  
  `ship_date` date,  
  `carrier` varchar(100),  
  `tracking_no` varchar(100),  
  `status` varchar(100),  
  PRIMARY KEY (`order_id`),  
  FOREIGN KEY (`order_id`)  
    REFERENCES `sales` (`order_id`),  
  FOREIGN KEY (`order_id`)
```

```

REFERENCES `customer_order` (`order_id`)
);

CREATE TABLE `department` (
  `department_id` integer,
  `department_name` varchar(50),
  PRIMARY KEY (`department_id`)
);

CREATE TABLE `employee` (
  `employee_id` integer,
  `employee_dpt` varchar(50),
  `employee_name` varchar(100),
  `start_date` date,
  `supervisor` varchar(100),
  PRIMARY KEY (`employee_id`),
  FOREIGN KEY (`employee_dpt`)
    REFERENCES `department` (`department_name`),
  FOREIGN KEY (`employee_id`)
    REFERENCES `sales` (`sales_id`)
);

```

Part III: Query Data

Question 1: Who are the top 3 salesperson with sales_id, what is their total sales and corresponding sales rank?

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```

12 --Question 1: Who are the top 3 salesperson with their sales_id, total sales amount and rank number?
13 select
14   sales_id,
15   employee_name as sales_full_name,
16   sum(total_amount) as total_sales,
17   rank() over (
18     order by sum(total_amount) desc
19   ) as sales_rank
20 from `smart_toys.sales` as s
21 left join `smart_toys.employee` as e
22 on s.sales_id = e.employee_id
23 group by sales_id, sales_full_name
24 order by sales_rank
25 limit 3;
26

```

✔ This script will process 2.03 KB when run.

Query results

Job information	Results	Chart	JSON	Execution details	Execution graph
Row	sales_id	sales_full_name	total_sales	sales_rank	
1	2137	Alice Hartin	159341.38	1	
2	2126	Emily Carter	115848.59	2	
3	2169	David Thompson	105959.95	3	

Question 2: What are the top 3 bestselling toys, their categories, and the total sales amount?

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```
11
12 --Question 2: Name the top 3 bestselling toys and list the total sales amount?
13
14 select
15   s.product_id,
16   t.category,
17   sum(s.total_amount) as total_sales
18 from `smart-toys.sales` as s
19 inner join `smart-toys.toys` as t
20 on s.product_id = t.toy_id
21 group by s.product_id, t.category
22 order by total_sales desc
23 limit 3;
```

24 This script will process 2.79 KB when run.

Query results

Job information Results Chart JSON Execution details Execution graph

Row	product_id	category	total_sales
1	gb010	STEM Toys	102523.24
2	gb009	Creative Toys	94484.12
3	gb007	Stuffed Animals	78344.07

Question 3: What is the average shipping duration for each carrier and total shipment?

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```
25
26 -- Question 3: List of average shipping duration in days for all carriers with least days on top.
27 SELECT
28   carrier,
29   ROUND(AVG(DATETIME_DIFF(DATETIME(ship_date), DATETIME(order_date), DAY)), 1) AS avg_ship_duration,
30   COUNT(*) AS total_shipments
31 FROM `smart-toys.shipment`
32 GROUP BY carrier
33 ORDER BY avg_ship_duration;
```

34 This script will process 2.79 KB when run.

Query results

Job information Results Chart JSON Execution details Execution graph

Row	carrier	avg_ship_duration	total_shipments
1	DHL	2.9	10
2	USPS	3.7	13
3	UPS	4.2	11
4	FedEx	4.6	16

Question 4: what are the total sales by month and show it in a line graph.

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```
1 --Question 4: what are the total sales by month and show it in a line graph.
2 SELECT
3   EXTRACT (Month
4   FROM
5   | transaction_date) AS month,
6   ROUND(SUM(total_amount),0) AS total_sales_by_month
7 FROM
8   `smart-toys.sales`
9 GROUP BY
10  month
11 ORDER BY
12  month;
```

Query completed

Query results

Job information Results Chart JSON Execution details Execution graph

month, total_sales_by_month by month

Month	Total Sales (by month)
1	45k
2	55k
3	55k
4	25k
5	35k
6	75k
7	65k
8	115k
9	75k
10	45k
11	35k
12	20k

Question5: What are the price ranges and product count for all toys?

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```
1 -- Question5: What are the price ranges and product count for all toys?
2 SELECT
3 CASE
4   WHEN unit_price <=10 THEN '< $10'
5   WHEN unit_price BETWEEN 10
6   AND 20 THEN '$10 to $20'
7   WHEN unit_price BETWEEN 20 AND 40 THEN '$20 to $40'
8   ELSE '>$40'
9 END
10 AS price_range,
11 COUNT(*) product_count
12 FROM
13   `smart_toys.toys`
14 GROUP BY
15   price_range;
```

✓ This query will process 80 B when run.

Query results

Job information Results Chart JSON Execution details Execution graph

Row	price_range	product_count
1	\$20 to \$40	4
2	\$10 to \$20	5
3	>\$40	1

Question6: What are the monthly sales and the cumulative monthly sales FOR the year 2023?

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Run Save query (Classic) Share Schedule Open in More

```
1 -- What are the monthly sales and the cumulative monthly sales FOR the year 2023?
2 WITH
3   cte_month AS (
4     SELECT
5       DATE_TRUNC(transaction_date, month) AS month_start,
6       ROUND(SUM(total_amount),0) AS monthly_sales
7     FROM
8       `smart_toys.sales`
9     GROUP BY
10      month_start )
11 SELECT
12   month_start,
13   monthly_sales,
14   SUM(monthly_sales) OVER (ORDER BY month_start ) AS monthly_cumulative_sales
15 FROM
16   cte_month
```

✓ This query will process 800 B when run.

Query results

Job information Results Chart JSON Execution details Execution graph

Row	month_start	monthly_sales	monthly_cumulat...
1	2023-01-01	50224.0	50224.0
2	2023-02-01	60091.0	110315.0
3	2023-03-01	59014.0	169329.0
4	2023-04-01	17211.0	186540.0
5	2023-05-01	30117.0	216657.0
6	2023-06-01	68384.0	285041.0
7	2023-07-01	58143.0	343184.0
8	2023-08-01	72856.0	416040.0
9	2023-09-01	113341.0	529381.0
10	2023-10-01	62669.0	592050.0